

THE VESTIBULAR SYSTEM

REQUIRED READINGS: Power Point Presentation, and Blumenfeld Chapter 12, page 523 – 525 & 527 - 529

LEARNING OBJECTIVES: after studying this chapter, students should be able to:

1. Discuss the function of the vestibular system
2. Describe the anatomy of the inner ear (vestibular apparatus)
3. Explain the location and difference in composition of the endolymph and perilymph
4. Explain the localization, morphology, and function of the vestibular receptors: the hair cells
5. Discuss the transduction mechanism in the vestibular system
6. Describe the type of head movement detected by the different receptors:
 - Angular acceleration – semi-circular canals
 - Linear acceleration – utricle and saccule
7. Explain the central vestibular pathways
 - The vestibular nuclei and their connections with cerebellum, spinal cord, and eye movement centers
 - The central pathways to the cerebral cortex
8. Explain the function and physiology of the vestibulo-ocular reflex
9. Apply the acquired knowledge to recognize peripheral and central disorders of the vestibular system
10. Describe the caloric test and its usefulness in clinical practice

Use this chapter as a guide for your textbook studies

OVERVIEW

The vestibular and auditory systems are the sensory systems associated with the vestibulocochlear nerve, cranial nerve VIII. The sensory receptors for both systems are in the inner ear in specialized sensory organs called the cochlea and vestibular apparatus. It is important to understand the basic anatomy of the inner ear and its specialized structures. This knowledge will be the basis to comprehend the function of the vestibular system. The information gathered and processed by the different vestibular structures is carried out through the vestibular pathways, which communicate with the spinal cord, the cerebellum, the cerebral cortex, and the ocular movement circuits to regulate posture, eye movements and eye position in response to movements of the head.

What follows is a guide for your readings; please refer to the abovementioned chapter in your textbook.

Comments on the Power Point Presentation

Slide 2: The study of the vestibular system starts with the understanding of its peripheral components in the inner ear: the vestibular apparatus (sometimes referred to as vestibular labyrinth,) with its bony and membranous parts. We need to remember the location, structure, and function of the 5-receptor sites: the utricle, the saccule, and the 3 semi-circular canals. It is important to understand the transduction mechanism at the hair cell level and finally to study the central vestibular pathway and its input to postural and eye movements.

The Vestibular System

1. The anatomy of the inner year
2. The vestibular apparatus
3. The vestibular receptors
4. The transduction mechanism
5. The central vestibular pathways

Slide 3: list the learning objective for this chapter

Learning Objectives

After studying these material you will be able to:

1. Discuss the function of the vestibular system
2. Describe the anatomy and histology of the vestibular apparatus
3. Explain the location and composition of the endolymph and perilymph
4. Describe the type of head movement detected by the different receptors:
 - Angular acceleration - semi-circular canals
 - Linear acceleration - utricle and saccule
5. Explain the localization, morphology and function of the vestibular receptors: the hair cells and discuss the transduction mechanism in the vestibular system
6. Explain the central vestibular pathways:
 - The vestibular nuclei and their connections with cerebellum, spinal cord, and eye movement centers
 - The central pathways to the cerebral cortex
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8. Describe the caloric testing
9. Apply the acquired knowledge to recognize peripheral and central disorders of the vestibular system

Slide 4: Vestibular System. This important system contributes greatly to the perception of self-motion, head position and spatial orientation relative to gravity. This information gives us the ability to stabilize gaze, head, and posture.



The Vestibular System

OBJ. # 1

Function:

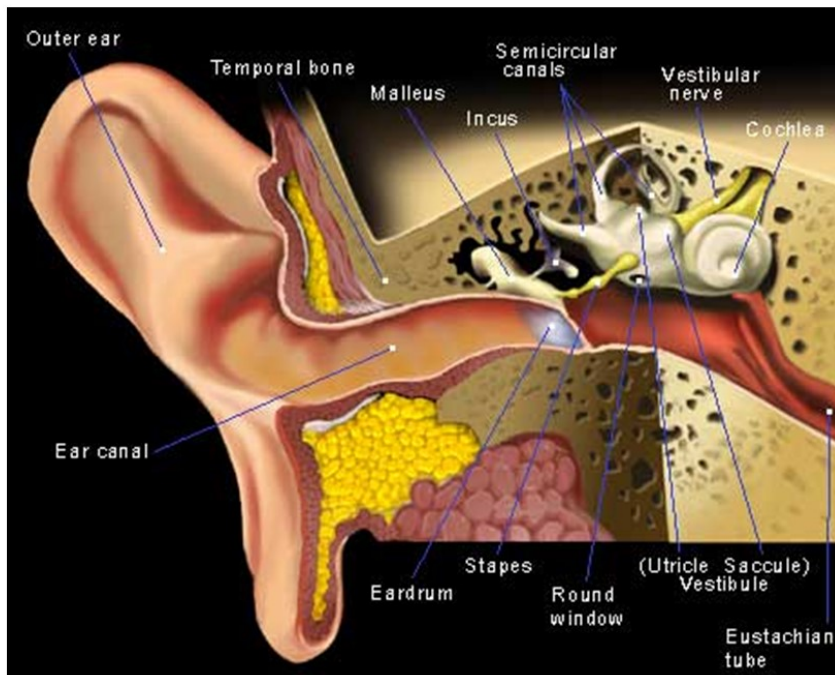
Provides information about head and body position in the space and allows for rapid compensatory responses to both self-induced and externally generated forces



Postural reflexes and eye movements



Slide 5: Diagram of the outer, middle, and inner ear. Today, we will learn about the anatomy and physiology of the vestibular apparatus located in the inner ear. The inner ear contains both the auditory and the vestibular peripheral receptors. Auditory receptors are located more anteriorly in the temporal bone in the cochlea. The vestibular receptors are located more posteriorly in a set of chambers named the vestibular labyrinth. The cochlea and the vestibular labyrinth are interconnected and the receptors in both cases are morphologically similar.

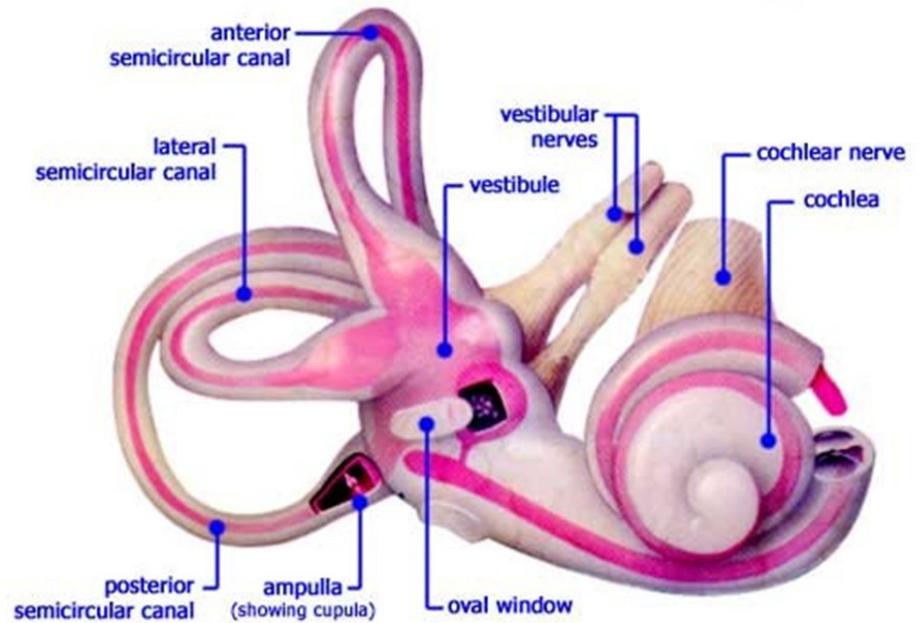


OBJ. # 2

Slide 6 - This slide shows the membranous labyrinth (pink structures). The vestibular apparatus consists of 2 large chambers called the utricle and saccule and the 3 semi-circular canals: anterior, posterior, and horizontal. Remember that like the cochlea, the space between the bone and the membranous labyrinth is filled with perilymph and the space inside of the membranous labyrinth is filled with endolymph. The main difference in composition between peri and endolymph is their potassium ion concentration with higher potassium ion in the endolymph.

The Inner Ear

OBJ. # 2 & 3

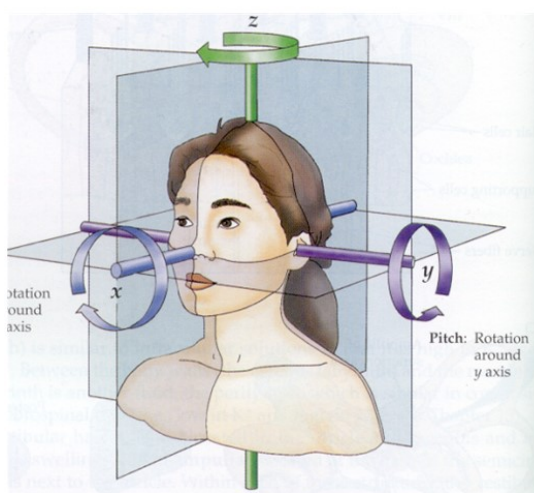


Slide 7 & 8- The head can move with six degrees of freedom: 3 rotational and 3 translational. The receptors for translational movement are in the dilated chambers of the vestibular *labyrinth*: the **otolith organs** in the utricle and saccule detect linear acceleration of the head: antero-posteriorly, sideways, and up-down, and also detect tilting of the head relative to gravity.

The receptors in the semi-circular canals are in the dilations of the *canals* (the part closer to the utricle) called **ampulla**. They are called **crista ampullaris** and detect angular acceleration of the head.

Head Movements Detected By The Vestibular Receptors

OBJ. # 4



Vestibular receptors - 5 different locations

At the crista ampullaris of the semi-circular canals:

- Anterior or superior
- Horizontal or lateral
- Posterior

Sense angular acceleration of the head (head rotation)

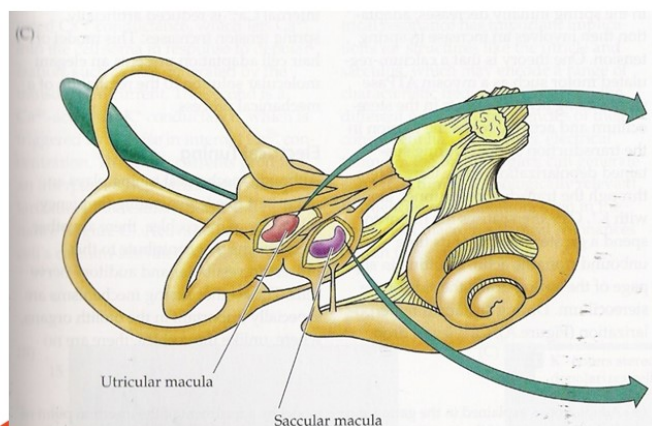
- The macula of the utricle and saccule

Sense linear acceleration (translation),
Static head position and head tilt relative to gravity

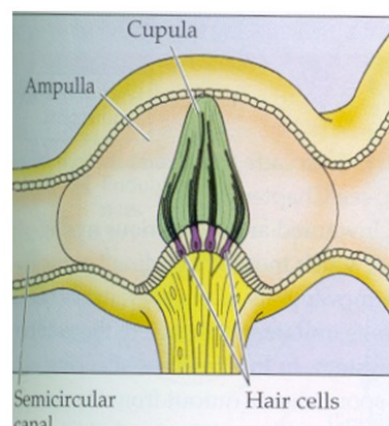
The Vestibular Receptors

OBJ. # 5

Utricle and saccule



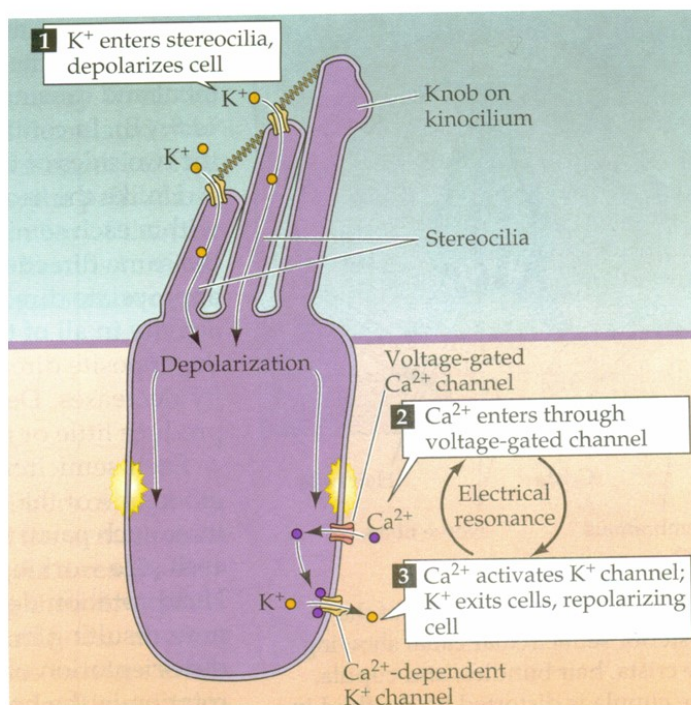
Semi-circular canals



Slide 9: This slide shows the structure of the hair cells and intends to remind you of the transduction mechanism that transforms the movements of the head into receptor potentials. The vestibular hair cells as their auditory counterparts are modified epithelial cells. The apical part of the cells is bathed by endolymph and contains stereocilia. The tallest stereocilia is called the kinocilium. The tips of these projections have *mechanically* gated potassium channels which open or close with the movements of the stereocilia. The kinocilium always faces the utricular side in the 3 semi-circular canals. Movements of the stereocilia towards the kinocilium (or towards the utricle) will open the channels on the tip of the stereocilia and potassium rushes into the cell from the potassium-rich endolymph. This depolarizes the hair cell and produces the release of neurotransmitter from the basal aspect of the cell where the nerve terminals are located. Depolarization also determines the opening of voltage-gated Ca^{2+} channels on the lateral aspects of the cell. As Ca^{2+} enters the cell, it triggers the opening of other potassium channels on the lateral and basal aspects of the cell membrane. Potassium then exits the cell thanks to the gradient produced by the low K^+ concentration of the perilymph that bathes the basal aspect of the hair cell.

Movement of the stereocilia away from the kinocilium closes the channels and then hyperpolarizes the receptor.

Hair cells tonically release neurotransmitter into the nerve fibers producing constant spontaneous activity in the vestibular nerve. Abnormalities of the hair cells or the transduction mechanism on one vestibular apparatus produce a decrease of the normal tonic activity of the receptors on that side which is perceived by the brain as an apparent increased activity on the contralateral normal side.



The Hair Cells And The Transduction Mechanism

OBJ. # 5

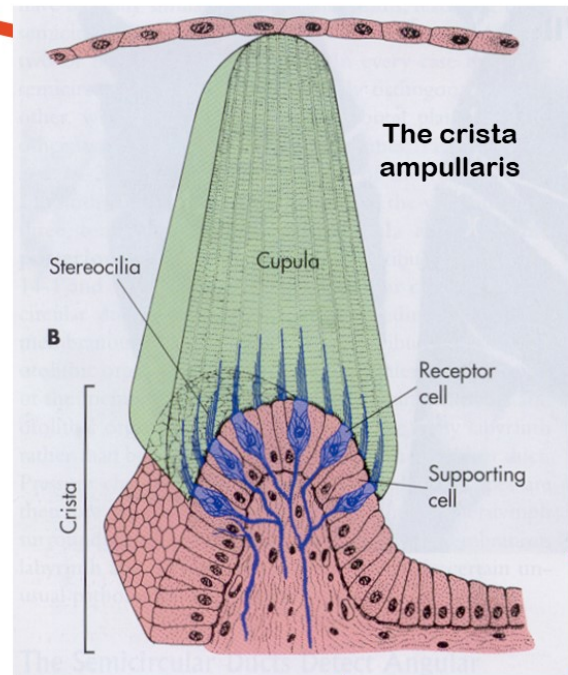
Slide 10: The **ampulla** in each semi-circular canal has a specialized sensory epithelium called the **crista ampullaris**, where the hair cells are located. The hair cell bundles are embedded in a gelatinous mass called the **cupula**, which extends the entire width of the ampulla, and the entire structure is bathed by endolymph. The cupula provides a barrier for the less dense endolymph and prevents it from freely circulating in the canal.

The Crista Ampullaris in the Semi-circular Canals

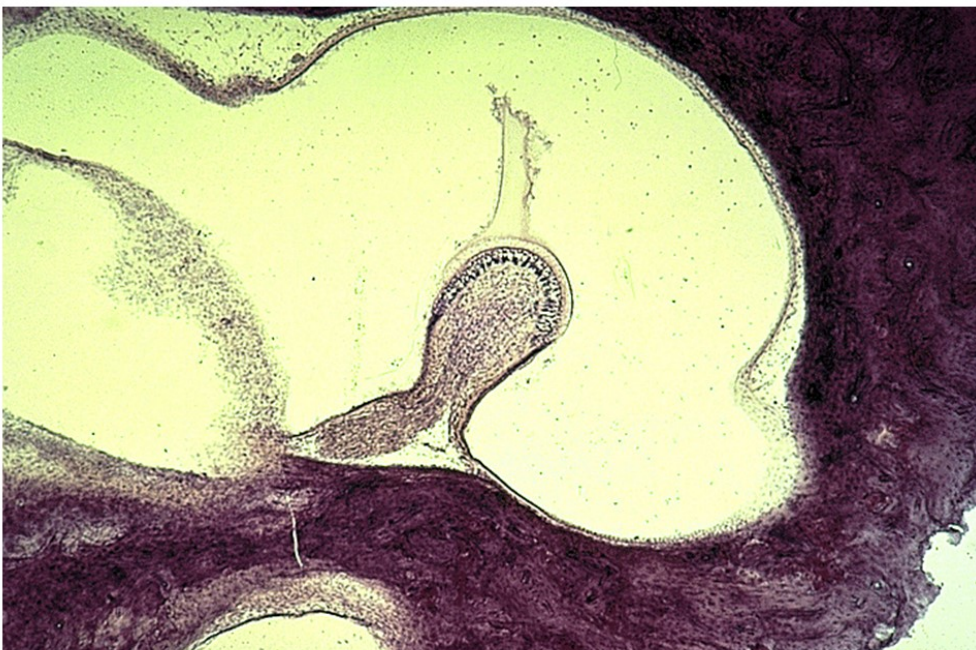
The hair cells of the semicircular canals are located in specialized dilations called ampullae

The sensory epithelium in each ampulla is called the crista, it contains the hair cells

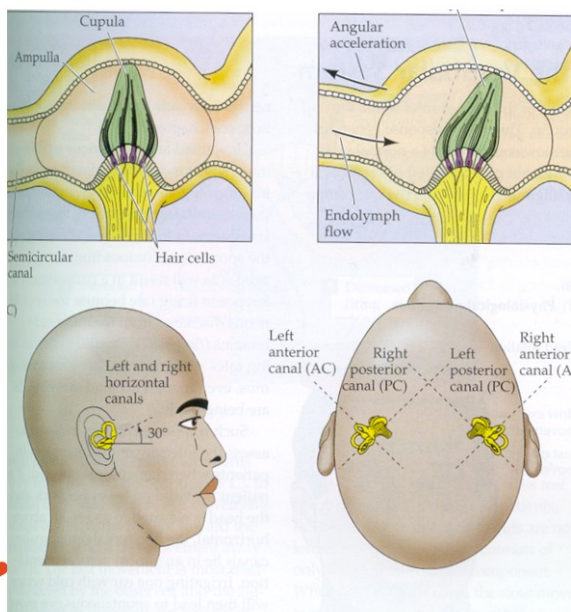
OBJ. # 5



Slide 11: Histological section of a semi-circular canal showing the crista ampullaris.



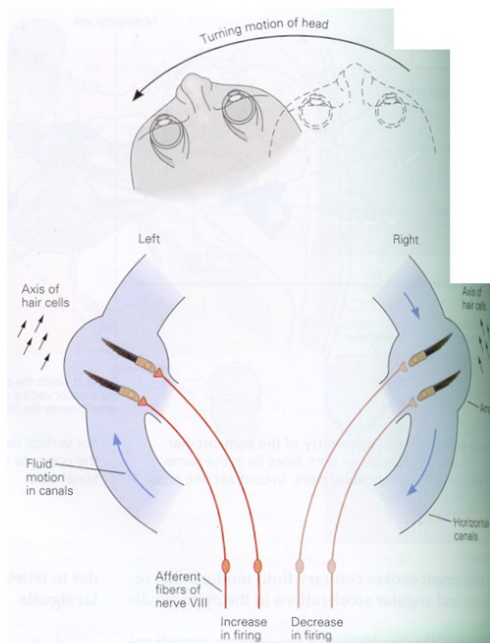
Slide 12 & 13: These diagrams show how the hair cells of the semi-circular canals transmit angular acceleration. When the head turns in the plane of one of the semi-circular canals, for simplicity let's say the horizontal plane and to the right, the inertia of the endolymph in the right ampulla pushes the cupula away from the direction of the movement, causing the displacement of the stereocilia towards the kinocilium. All the hair cells in the horizontal canal (in our example) will behave the same way, at the same time opening potassium channels and depolarizing the cells. Notice that all the receptors have the kinocilium located towards the utricle in the crista ampullaris, so that all the cells depolarize with the same type of movement. It is important to realize that each semi-circular canal works together with its opposite partner on the other side of the head. In our example the right horizontal semi-circular canal depolarizes while the left horizontal canal hyperpolarizes signaling the brain the orientation of the head movement.



The Semi-circular Canals

How the Semi-circular canals detect movement?

OBJ. # 5



Angular acceleration of the head to the left produces movement of the endolymph to the right.

Deflection of the stereocilia towards the utricle depolarizes the hair cells on the left. The opposite occurs on the right side with hyperpolarization of the hair cells

OBJ. # 5

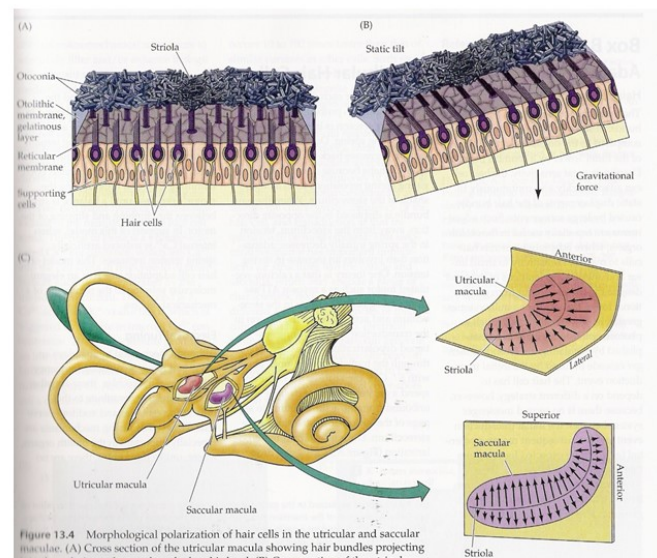
Slide 14 & 15 - The other 2 receptor locations are in the otolith organs of the utricle and saccule which detect linear acceleration of the head and tilting of the head against gravity. The **otolith organs** consist of a sensory epithelium called the **macula** where the hair cells are located. The macula is a kidney-shaped structure oriented vertically in the saccule and horizontally in the utricle. Each macula is divided into 2 parts by a specialized central structure called the **striola**. The striola divides the hair cells into 2 populations with different polarities. This means that the kinocilium of these cells are organized in all the possible orientations around the striola, so that the cells detect movement of the head in all the possible directions. A gelatinous layer called the **otolith membrane** covers the hair cells of the macula. The hair cells have their stereocilia and kinocilium embedded in it. On top of the otolith membrane is a fibrous structure that contains multiple crystals of calcium carbonate called **otoconia** making the entire structure heavier than the surrounding endolymph.

Linear acceleration of the head in a particular direction produces displacement of the otoconia relative to the gelatinous membrane where the stereocilia are embedded. The stereocilia aligned with the appropriate head movement turn towards the kinocilium and the cell depolarizes. Looking at the orientation of the hair cells in the otolith organs, it is evident that during any given movement some cells will be depolarized indicating the direction of the movement, while other cells will be hyperpolarized in the same macula because that movement does not activate them. Each macula works together with its partner on the opposite side of the head, so any displacement of the head produces opposite effects on both sides signaling to the brain the direction of the movement.

The Vestibular Receptors

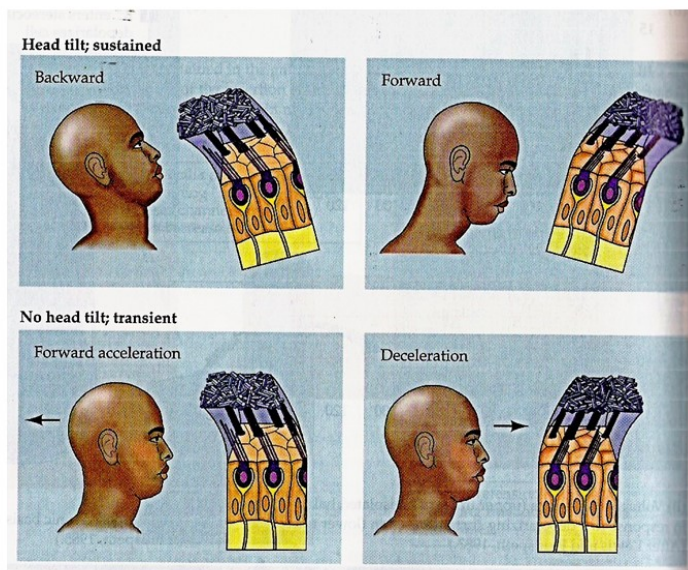
The hair cell bundles in each vestibular organ have specific orientations

The utricle and saccule contain a sensory epithelium called the macula



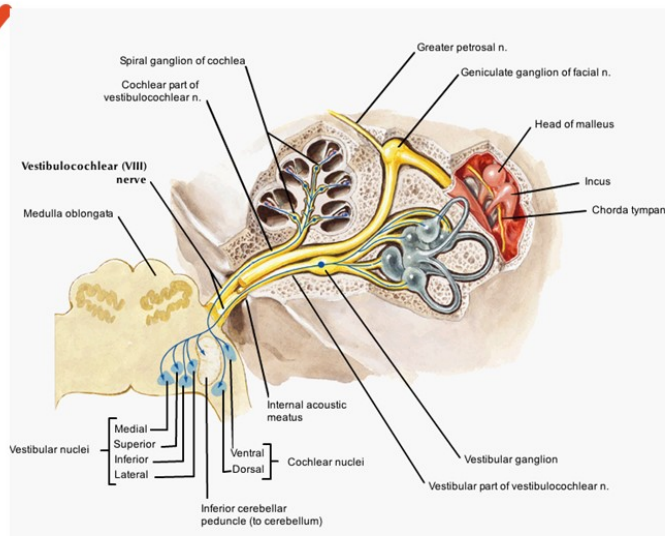
Linear acceleration

The Vestibular Receptors



Slide 16: The central vestibular pathways - This diagram taken from Netter's atlas shows the temporal bone and the inner ear. We can see the cochlea and the vestibular labyrinth. We also see the vestibulocochlear nerve and the facial nerve traveling in the internal acoustic meatus. The vestibular ganglion houses the **bipolar neurons**, the first neurons of the vestibular pathway. The peripheral projection of these cells contact the base of the hair cells and the central projection takes the information to the vestibular nuclei in the brainstem and to the flocculonodular lobe of the cerebellum (primary vestibular fibers). These fibers travel in the **inferior cerebellar peduncle**.

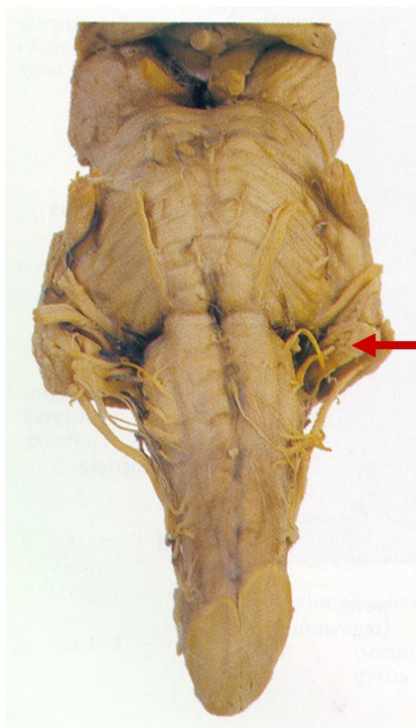
OBJ. # 6



The Vestibular Pathways

The vestibular receptors are contacted by the primary afferent fiber

Slide 17: Shows the ventral surface of the brainstem to see the CN VIII as it enters at the pontocerebellar angle.



CN VIII
Vestibulocochlear
at pontocerebellar
angle

Ventral Surface Of The Brainstem

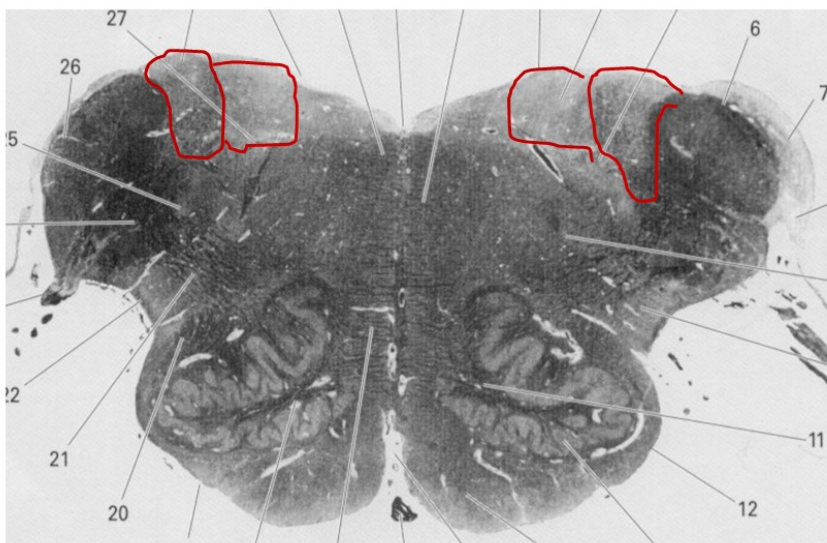
OBJ. # 6

Slide 18: This is a histological staining of the rostral medulla where the vestibular nuclei and the inferior cerebellar peduncles can be observed.

Rostral Medulla

Location of 2 Vestibular nuclei on the floor of the IV ventricle

OBJ. # 6



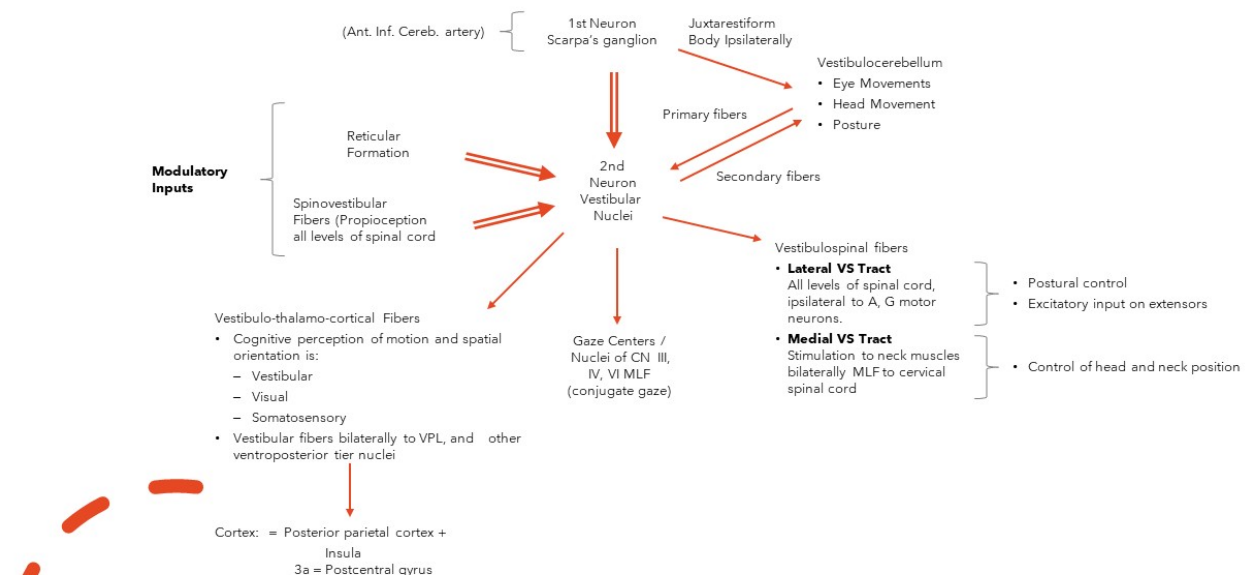
Slide 19: This is a diagram showing the entire vestibular connections. It starts with the **first neuron of the pathways in the vestibular (Scarpa's) ganglion**. The second neuron of the pathway is located either in the vestibular nuclei or in the flocculonodular lobe of the cerebellum. The vestibular nuclei receive modulatory inputs from the reticular formation, the spinal cord, the cerebellar cortex, and the vestibular nuclei on the contralateral side of the brainstem. It is important to remember that the vestibulo-cerebellar communication is bidirectional.

After processing the input information, the vestibular nuclei project to:

- The spinal cord through the medial and lateral vestibulospinal tracts
- Thalamocortical fibers – to the VPL and all posterior nuclei of the thalamus. The thalamus projects in turn to several cortical areas
- To the gaze centers in the pons and midbrain, and the nuclei of CNs III, IV, and VI – to coordinate eye movements, and the Vestibulo-Ocular Reflex through the ascending MLF

The Central Vestibular Pathways

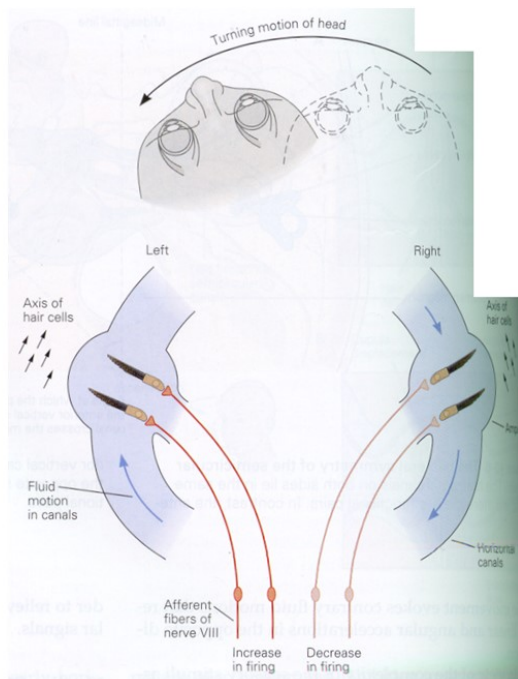
OBJ. # 6



Slide 20 - The Vestibulo-ocular reflex – The function of the VOR is to stabilize the gaze when the head is moving to keep an object of interest in the center of the fovea. It is triggered by the movement of the head when fixating on an object of interest which is *stationary*. There are 3 vestibulo-ocular reflexes:

- Rotational VOR compensates for head rotation
- Translational VOR compensates for linear head movements
- Ocular counter-rolling response compensates for head tilt in the vertical plane

As you fixate an object in your visual field (try it yourself) and at the same time you move away from the object (to the right or left), your eyes will move the same amount and in the opposite direction of your head movement. (Try this in a mirror, watch your eyes.) This movement of the eyes is reflexive and controlled by the vestibular system and some brainstem centers.



The Vestibulo-Ocular Reflex

- As the head rotates toward one side, the eyes will initiate a slow reflexive movement to the contralateral side to keep the object of interest in the center of the visual field

OBJ. # 7

Slide 21 & 22 – These slides show how this reflex is controlled. On the left diagram in slide 21 (excitatory pathways) the left horizontal canal is stimulated by the movement of the head towards the left. The left head movement makes the endolymph of the left horizontal canal move to the right due to inertia. The endolymphatic movement pushes the stereocilia of the crista ampullaris towards the kinocilium, a movement that results in the depolarization of the hair cells on the left. In the right horizontal canal, the opposite occurs making the hair cells to hyperpolarize.

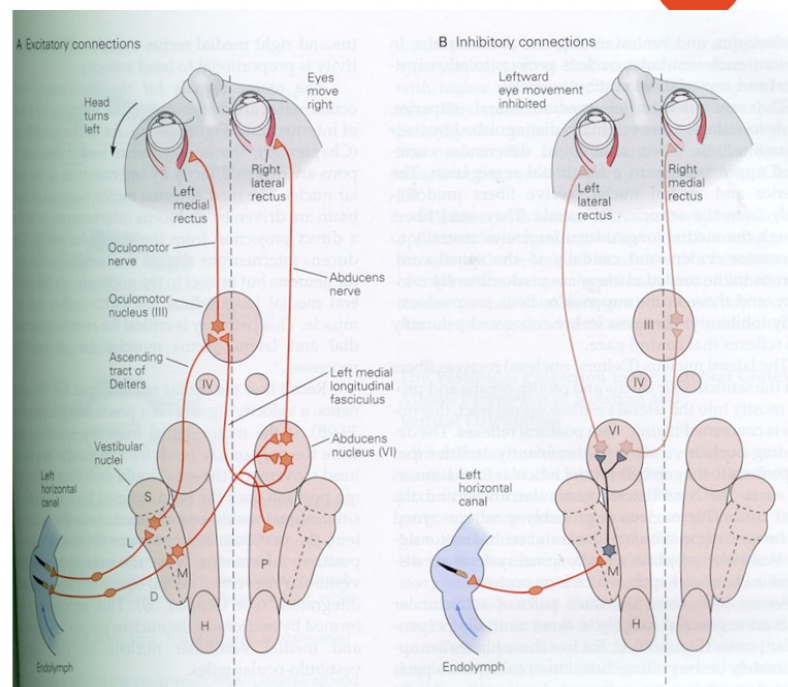
The primary left afferent fibers in the vestibular nerve send the excitatory information to the vestibular nuclei. The diagram shows how 2 different vestibular nuclei neurons receive this information. One of them communicates with the abducens nucleus contralaterally to activate the right lateral rectus. The right abducens nucleus communicates with the left oculomotor nucleus via MLF to activate the left medial rectus. A second neuron in the left vestibular nuclei receives information from the primary afferents; this neuron sends a redundant excitatory communication to the left oculomotor (ipsilateral) nucleus for the contraction of the left medial rectus. Both eyes move slowly to the right (slow phase of the reflex). Information about movement of the head to the left reaches also the right vestibular nuclei.

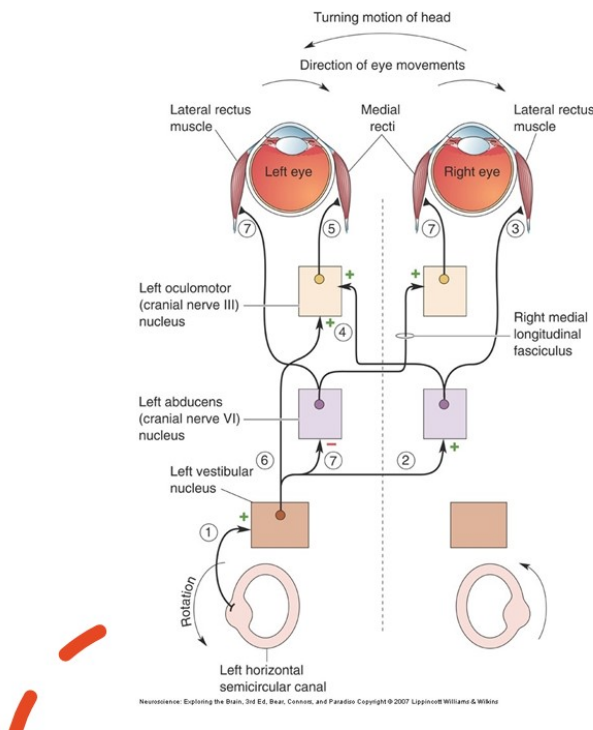
On the right-hand side diagram, we see how one neuron in the left vestibular nuclei (inhibitory interneuron) sends an inhibitory signal to the ipsilateral abducens nucleus so that the left lateral rectus is inhibited and at the same time communicates with the right oculomotor nucleus to inhibit the right medial rectus. These inhibitions favor the coordinated eye movements to the right.

Testing the VOR is important in unconscious patients when we want to know if the eye movement machinery in the brainstem is intact. To do that, we use the doll's eyes maneuver or the caloric testing.

The Vestibulo-Ocular REFLEX

OBJ. # 7





Coordination of the VOR by the vestibular system triggered by head movement

OBJ. # 7

Slide 23 reminds us of the main symptoms and signs present in patients with lesions of the vestibular system or the inferior cerebellar peduncles.

Clinical Correlations

Lesions of the vestibular pathways or the fibers of the inferior cerebellar peduncle produces 2 main symptoms:

- Pathological Nystagmus and
- Vertigo

Other symptoms and signs that could be associated are:

- Nausea and vomiting
- Hearing loss
- Wide-based gait
- Unsteadiness

Slide 24 explains the Caloric Testing. The vestibular system projects to the nuclei of III, IV and VI to always control the position of the eyes. This constant tonic input keeps all the extrinsic muscles of the eyes with the appropriate tension to maintain our eyes in primary gaze in the center of the orbit. Pathological nystagmus occurs when there is an imbalance between one vestibular apparatus with respect to the other. When one side of the vestibular pathways is damaged either in the periphery or in the CNS, its tonic output is depressed or absent, leaving only unopposed activity from the contralateral side. The effect of this is illusory perception of head movement.

In the case of vestibular neuritis, the posterior and anterior canal output is mostly cancelled out, leaving mainly the depressed output from the horizontal canal unopposed by the contralateral horizontal canal. The net effect is a drift of the eyes towards the effected side with a corrective saccade in the opposite direction.

Nystagmus is always named by the direction of the saccadic eye movement.

It is sometimes useful to assess the function of this system more directly than with physical exam. In this case you can use the **Caloric Testing**. Caloric testing consists of irrigating the ears with either cold or warm water. The patient should be in a supine position with head tilted forward 30 degrees. Each external auditory canal is irrigated for 30 seconds first with water at degrees C, and then at 44 C with a 5 minutes interval between each irrigation. In a conscious person with a normally functioning vestibular system this irrigation triggers the following eye movements:

Cold – contralateral nystagmus (slow drift towards, saccadic correction away)

Warm – ipsilateral nystagmus (slow drift away, saccadic correction towards)

Since nystagmus is named for the saccadic movement, you can remember the mnemonic COWS: cold-opposite, warm-same. Be careful to remember that the saccadic component of nystagmus requires fixation. In other words, the patient must have visual acuity sufficient to fixate on a target and be conscious. Outside of specialty situations, calorics are frequently used to assess brainstem function in *unconscious* patients as part of a brain death evaluation. In this situation, only the slow movement is observed and the COWS mnemonic fails.

Clinical Correlations

Pathological Nystagmus - Caloric Testing of vestibular function

* **Mnemonic:** COWS - cold=opposite, warm=same

Irrigation of the ear with cold water

produces slow movements of the eyes to the ipsilateral side because convection currents mimic head movement away to the irrigated side

Irrigation of the ear with warm water

produces slow movements of the eyes to the contralateral side because convection currents mimic head movement to the irrigated side

